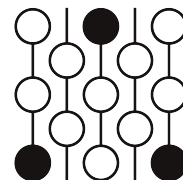


Review of Submission 4B

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Instead of the theorem in the problem, this submission wrote down a different theorem and then proved it. This different theorem follows from standard facts in a fairly easy way. It does not require the main mathematical ideas in the solution of the actual problem.

The submission does correctly explain some standard facts that are relevant to the problem. But it does not make real progress on the question asked.

Correctness. The submission states a different theorem than the one in the problem. The proof of the different theorem is correct. There is one place where there is either a small error or some bad writing, but it can be fixed easily.

There are also some issues with the organization of the writing. The statement of the theorem is very long, and some of the statement should really be moved to the proof. Some parts of the argument are never used and could be omitted. On the other hand, on a line-by-line basis, the writing is clear.

Novelty. I don't see anything novel in the proof. The ingredients are standard.

2 Recommendation

The paper should be rejected. It does correctly explain some standard facts in the general area of the problem. But it does not make real progress on the question asked.

3 Submission with inline comments

Theorem (fiber-filling and coarea reduction)

Let

$$R = \prod_{i=1}^4 [0, R_i], \quad S = \prod_{i=1}^4 [0, S_i],$$

with $0 < R_1 \leq R_2 \leq R_3 \leq R_4$ and $0 < S_1 \leq S_2 \leq S_3 \leq S_4$. Orient S by

$$dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4.$$

Write

$$Q = [0, S_1] \times [0, S_2], \quad B = [0, S_3] \times [0, S_4],$$

so $S = Q \times B$, and orient Q and B by $dx_1 \wedge dx_2$ and $dx_3 \wedge dx_4$, respectively. Let

$$\pi : S \rightarrow B, \quad \pi(x_1, x_2, x_3, x_4) = (x_3, x_4),$$

and suppose

$$f : (R, \partial R) \rightarrow (S, \partial S)$$

is a Lipschitz finite piecewise smooth map of degree 1 with $\text{Dil}_2(f) \leq 1$. Put

$$F = \pi \circ f = (f_3, f_4), \quad G = (f_1, f_2).$$

For $y \in B$, set

$$P_y = Q \times \{y\}.$$

Use the slicing convention normalized by

$$\langle \llbracket Q \times B \rrbracket, \pi, y \rangle = \llbracket Q \times \{y\} \rrbracket.$$

Then for a.e. $y \in \text{int } B$, the slice

$$z_y = \langle \llbracket R \rrbracket, F, y \rangle$$

is a relative integral 2-cycle in $(R, \partial R)$, and

$$f_{\#} z_y = \llbracket P_y \rrbracket.$$

Consequently,

$$\text{FillVol}_R(z_y) \geq \text{FillVol}_S(P_y) = S_1 S_2 \text{dist}(y, \partial B).$$

Moreover,

$$\int_B \text{Area}(z_y) dy \leq \text{Vol}(R).$$

Finally, let

$$B_0 = \left[\frac{S_3}{3}, \frac{2S_3}{3} \right] \times \left[\frac{S_4}{3}, \frac{2S_4}{3} \right].$$

At points where dF has rank 2, let

$$\sigma_1(x) \geq \sigma_2(x) > 0$$

be the two nonzero singular values of dF_x . Define the measure

$$d\mu = \frac{\sigma_2}{\sigma_1} dx$$

on the rank-two set of dF , and $d\mu = 0$ where $\text{rank } dF < 2$. Then

$$\mu(F^{-1}(B_0)) \geq \frac{1}{9} S_1 S_2 S_3 S_4$$

and

$$\int_{B_0} \text{Area}(z_y) dy = \int_{F^{-1}(B_0)} \sigma_1^2 d\mu.$$

Thus, if in addition one has an estimate

$$\int_{F^{-1}(B_0)} \sigma_1^2 d\mu \geq c_d \sqrt{\frac{S_3}{S_2}} \mu(F^{-1}(B_0))$$

[Reviewer comment: This last hypothesis is not like anything in the original problem. Adding this hypothesis makes the conclusion more-or-less immediate.]

for some constant $c_d > 0$, then

$$R_1 R_2 R_3 R_4 \geq \frac{c_d}{9} S_1 S_2^{1/2} S_3^{3/2} S_4.$$

Proof

Write $\mathbf{M}(T)$ for the mass of an integral current T , and write $\llbracket X \rrbracket$ for the integral current associated to an oriented Lipschitz manifold or rectangle X .

For a compact rectangle X , if $Z \in \mathbf{I}_m(X)$ satisfies $\text{spt } \partial Z \subset \partial X$, define its relative filling volume by

$$\text{FillVol}_X(Z) = \inf \{ \mathbf{M}(Y) : Y \in \mathbf{I}_{m+1}(X), \text{spt}(\partial Y - Z) \subset \partial X \}.$$

We use the standard slicing theorem for integral currents: if $T \in \mathbf{I}_m$ and Φ is Lipschitz into $\mathbb{R}^k$, then for a.e. a , the slice $\langle T, \Phi, a \rangle \in \mathbf{I}_{m-k}$ exists and

$$\partial \langle T, \Phi, a \rangle = \pm \langle \partial T, \Phi, a \rangle.$$

We also use slicing naturality: for Lipschitz H ,

$$H_{\#} \langle T, \Phi \circ H, a \rangle = \langle H_{\#} T, \Phi, a \rangle$$

for a.e. a . Finally, we use the rectifiable-current area inequality

$$\mathbf{M}(H_{\#} T) \leq \int J_k(dH|_{\text{Tan } T}) d\|T\|$$

for $T \in \mathbf{I}_k$.

First prove the fundamental current identity

$$f_{\#} \llbracket R \rrbracket = \llbracket S \rrbracket.$$

Let

$$T = f_{\#} \llbracket R \rrbracket.$$

Since f is Lipschitz, $T \in \mathbf{I}_4(S)$, and

$$\partial T = f_{\#} \partial \llbracket R \rrbracket$$

is supported in $f(\partial R) \subset \partial S$. Hence, in $\text{int } S$,

$$\partial(T \llcorner \text{int } S) = 0.$$

By the constancy theorem on the connected open set $\text{int } S$,

$$T \llcorner \text{int } S = m \llbracket \text{int } S \rrbracket$$

for some $m \in \mathbb{Z}$. Therefore $T - m \llbracket S \rrbracket$ is supported in ∂S . Since ∂S has $\mathcal{H}^4$ -measure zero, no nonzero integral 4-current can be supported there, so

$$T = m \llbracket S \rrbracket.$$

Because f has degree 1, T represents the relative fundamental class $[S, \partial S]$, hence $m = 1$. Thus

$$f_{\#} \llbracket R \rrbracket = \llbracket S \rrbracket.$$

Now apply slicing to $F = \pi \circ f$. For a.e. $y \in \text{int } B$,

$$z_y = \langle \llbracket R \rrbracket, F, y \rangle$$

is an integral 2-current, and

$$\partial z_y = \pm \langle \partial \llbracket R \rrbracket, F, y \rangle,$$

so ∂z_y is supported in ∂R . Hence z_y is a relative integral 2-cycle in $(R, \partial R)$.

By slicing naturality,

$$f_{\#} z_y = f_{\#} \langle \llbracket R \rrbracket, F, y \rangle = \langle f_{\#} \llbracket R \rrbracket, \pi, y \rangle = \langle \llbracket S \rrbracket, \pi, y \rangle.$$

With the stated product orientation and slice convention,

$$\langle \llbracket S \rrbracket, \pi, y \rangle = \llbracket Q \times \{y\} \rrbracket = \llbracket P_y \rrbracket.$$

Thus

$$f_{\#} z_y = \llbracket P_y \rrbracket.$$

Next, $\text{Dil}_2(f) \leq 1$ implies $\text{Dil}_3(f) \leq 1$. Indeed, at a differentiability point of f , let

$$a_1 \geq a_2 \geq a_3 \geq a_4 \geq 0$$

be the singular values of df . The hypothesis gives $a_1 a_2 \leq 1$. Hence $a_2 \leq 1$, so $a_3 \leq 1$, and therefore

$$a_1 a_2 a_3 \leq a_1 a_2 \leq 1.$$

Thus $\|\Lambda^3 df\| \leq 1$ a.e.

Let $Y \in \mathbf{I}_3(R)$ be a relative filling of z_y , so

$$\text{spt}(\partial Y - z_y) \subset \partial R.$$

Then

$$\partial(f_{\#}Y) - f_{\#}z_y = f_{\#}(\partial Y - z_y)$$

is supported in $f(\partial R) \subset \partial S$. Since $f_{\#}z_y = \llbracket P_y \rrbracket$, the current $f_{\#}Y$ is a relative filling of P_y in S . Also,

$$\mathbf{M}(f_{\#}Y) \leq \int \|\Lambda^3 df\| d\|Y\| \leq \mathbf{M}(Y).$$

Taking the infimum over all such Y gives

$$\text{FillVol}_R(z_y) \geq \text{FillVol}_S(P_y).$$

It remains to compute $\text{FillVol}_S(P_y)$. Let

$$\delta(y) = \text{dist}(y, \partial B).$$

For the upper bound, choose a shortest segment $\gamma \subset B$ from ∂B to y , oriented toward y . Then

$$C = \llbracket Q \rrbracket \times \llbracket \gamma \rrbracket$$

is a relative filling of P_y in S , and

$$\mathbf{M}(C) = S_1 S_2 \delta(y).$$

Hence

$$\text{FillVol}_S(P_y) \leq S_1 S_2 \delta(y).$$

For the lower bound, define

$$\delta(q) = \text{dist}(q, \partial B) = \min\{q_3, S_3 - q_3, q_4, S_4 - q_4\}.$$

For $\eta > 0$, set

$$m_{\eta}(q) = -\eta \log \left(e^{-q_3/\eta} + e^{-(S_3-q_3)/\eta} + e^{-q_4/\eta} + e^{-(S_4-q_4)/\eta} \right).$$

Then

$$\delta(q) - \eta \log 4 \leq m_{\eta}(q) \leq \delta(q), \quad |\nabla m_{\eta}(q)| \leq 1.$$

Choose a smooth function $\chi_{\eta} : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\chi_{\eta}(t) = 0 \quad \text{for } t \leq 0, \quad 0 \leq \chi'_{\eta}(t) \leq 1, \quad \chi_{\eta}(t) \geq t - \eta \quad \text{for } t \geq 0.$$

Set

$$\rho_{\eta}(q) = \chi_{\eta}(m_{\eta}(q)).$$

Then ρ_η is smooth,

$$\rho_\eta = 0 \quad \text{on } \partial B, \quad |d\rho_\eta| \leq 1,$$

and for every fixed $y \in \text{int } B$,

$$\rho_\eta(y) \rightarrow \delta(y) \quad \text{as } \eta \rightarrow 0.$$

Define the smooth 2-form

$$\alpha_\eta = \rho_\eta(x_3, x_4) dx_1 \wedge dx_2.$$

It restricts to zero on ∂S : on the x_1 - and x_2 -faces because $dx_1 \wedge dx_2$ vanishes there, and on the x_3 - and x_4 -faces because $\rho_\eta = 0$ on ∂B .

Let $Y \in \mathbf{I}_3(S)$ be any relative filling of P_y , so

$$\text{spt}(\partial Y - \llbracket P_y \rrbracket) \subset \partial S.$$

Since α_η vanishes on ∂S ,

$$\partial Y(\alpha_\eta) = \llbracket P_y \rrbracket(\alpha_\eta) = \rho_\eta(y) S_1 S_2.$$

By Stokes' theorem for currents,

$$Y(d\alpha_\eta) = \rho_\eta(y) S_1 S_2.$$

But

$$d\alpha_\eta = d\rho_\eta \wedge dx_1 \wedge dx_2,$$

whose comass is at most $|d\rho_\eta| \leq 1$. Therefore

$$\mathbf{M}(Y) \geq \rho_\eta(y) S_1 S_2.$$

Letting $\eta \rightarrow 0$ gives

$$\mathbf{M}(Y) \geq S_1 S_2 \delta(y).$$

Taking the infimum over Y ,

$$\text{FillVol}_S(P_y) \geq S_1 S_2 \delta(y).$$

Thus

$$\text{FillVol}_S(P_y) = S_1 S_2 \text{dist}(y, \partial B).$$

In particular, for $y \in B_0$,

$$\text{dist}(y, \partial B) \geq \frac{S_3}{3},$$

because $S_4 \geq S_3$. Hence

$$\text{FillVol}_R(z_y) \geq \frac{1}{3} S_1 S_2 S_3.$$

Now apply the coarea formula to the Lipschitz map $F : R \rightarrow B$. For every nonnegative Borel function g ,

$$\int_B \int_{F^{-1}(y)} g d\mathcal{H}^2 dy = \int_R g J_2 F dx,$$

where

$$J_2 F = \|\Lambda^2 dF\|.$$

For a.e. y , the mass of the slice z_y is $\text{Area}(z_y)$, so with $g = 1$,

$$\int_B \text{Area}(z_y) dy = \int_R J_2 F dx.$$

Since $F = \pi \circ f$ and π is an orthogonal projection,

$$J_2 F \leq \|\Lambda^2 df\| \leq 1.$$

Therefore

$$\int_B \text{Area}(z_y) dy \leq \text{Vol}(R).$$

It remains to prove the two estimates involving μ . Fix a regular value $y \in B_0$, and let $x \in z_y$ be a point where f is differentiable and dF_x has rank 2. Then

$$T_x z_y = \ker dF_x.$$

Let

$$N_x = (T_x z_y)^\perp.$$

The restriction $dF_x|_{N_x} : N_x \rightarrow \mathbb{R}^2$ has singular values $\sigma_1(x) \geq \sigma_2(x) > 0$.

Choose a unit vector $n \in N_x$ such that

$$|dF_x(n)| = \sigma_1(x).$$

For any unit vector $v \in T_x z_y$, one has $dF_x(v) = 0$. Since

$$df_x = (dG_x, dF_x),$$

we get

$$df_x(v) = (dG_x(v), 0), \quad df_x(n) = (dG_x(n), dF_x(n)).$$

In the orthogonal splitting $\mathbb{R}^4 = \mathbb{R}_Q^2 \oplus \mathbb{R}_B^2$, the mixed $Q \wedge B$ component of

$$df_x(v) \wedge df_x(n)$$

is

$$dG_x(v) \wedge dF_x(n),$$

whose norm is

$$|dG_x(v)| |dF_x(n)| = |dG_x(v)| \sigma_1(x).$$

This mixed component has norm at most the norm of the full wedge. Therefore, using $\text{Dil}_2(f) \leq 1$,

$$|dG_x(v)| \sigma_1(x) \leq |df_x(v) \wedge df_x(n)| \leq 1.$$

Hence

$$|dG_x(v)| \leq \sigma_1(x)^{-1}.$$

Taking the supremum over unit $v \in T_x z_y$,

$$\|dG_x|_{T_x z_y}\| \leq \sigma_1(x)^{-1}.$$

Since $T_x z_y$ is two-dimensional,

$$J_2(G|_{T_x z_y}) \leq \sigma_1(x)^{-2}.$$

Let $p_Q : S \rightarrow Q$ be the projection onto the first two coordinates. Since $G = p_Q \circ f$, for a.e. $y \in B_0$,

$$G_{\#} z_y = (p_Q)_{\#} f_{\#} z_y = (p_Q)_{\#} [P_y] = [Q].$$

Therefore

$$S_1 S_2 = \mathbf{M}([Q]) = \mathbf{M}(G_{\#} z_y) \leq \int_{z_y} J_2(G|_{T z_y}) dA \leq \int_{z_y} \sigma_1^{-2} dA.$$

Integrating over B_0 ,

$$\int_{B_0} \int_{z_y} \sigma_1^{-2} dA dy \geq S_1 S_2 |B_0| = \frac{1}{9} S_1 S_2 S_3 S_4.$$

Now define

$$\tilde{g}(x) = \sigma_1(x)^{-2} \mathbf{1}_{F^{-1}(B_0) \cap \{\text{rank } dF=2\}}(x).$$

For $M > 0$, set $g_M = \min\{\tilde{g}, M\}$. By the coarea formula,

$$\int_{B_0} \int_{z_y} g_M dA dy = \int_{F^{-1}(B_0)} g_M J_2 F dx.$$

Letting $M \rightarrow \infty$ and using monotone convergence,

$$\int_{B_0} \int_{z_y} \sigma_1^{-2} dA dy = \int_{F^{-1}(B_0) \cap \{\text{rank } dF=2\}} \sigma_1^{-2} J_2 F dx.$$

On the rank-two set,

$$J_2 F = \sigma_1 \sigma_2.$$

Thus

$$\sigma_1^{-2} J_2 F = \frac{\sigma_2}{\sigma_1}.$$

Hence

$$\int_{B_0} \int_{z_y} \sigma_1^{-2} dA dy = \int_{F^{-1}(B_0)} \frac{\sigma_2}{\sigma_1} dx = \mu(F^{-1}(B_0)).$$

Combining this with the previous inequality gives

$$\mu(F^{-1}(B_0)) \geq \frac{1}{9} S_1 S_2 S_3 S_4.$$

Finally, applying the ordinary coarea formula over B_0 ,

$$\int_{B_0} \text{Area}(z_y) dy = \int_{F^{-1}(B_0)} J_2 F dx.$$

On the rank-two set,

$$J_2 F = \sigma_1 \sigma_2 = \sigma_1^2 \frac{\sigma_2}{\sigma_1},$$

while both sides vanish on the rank < 2 set. Therefore

$$\int_{B_0} \text{Area}(z_y) dy = \int_{F^{-1}(B_0)} \sigma_1^2 d\mu.$$

Assume now that

$$\int_{F^{-1}(B_0)} \sigma_1^2 d\mu \geq c_d \sqrt{\frac{S_3}{S_2}} \mu(F^{-1}(B_0)).$$

Then

$$\int_{B_0} \text{Area}(z_y) dy \geq c_d \sqrt{\frac{S_3}{S_2}} \mu(F^{-1}(B_0)) \geq \frac{c_d}{9} S_1 S_2 S_3 S_4 \sqrt{\frac{S_3}{S_2}}.$$

Since

$$\text{Vol}(R) \geq \int_B \text{Area}(z_y) dy \geq \int_{B_0} \text{Area}(z_y) dy,$$

we obtain

$$R_1 R_2 R_3 R_4 = \text{Vol}(R) \geq \frac{c_d}{9} S_1 S_2^{1/2} S_3^{3/2} S_4.$$

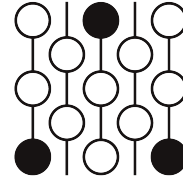
This completes the proof.

Review of Submission 4B

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. This is a little difficult to read - the statement of the “Theorem” is spread out over two pages and makes a lot of disconnected claims.

The key part of the statement is:

[I]f in addition one has an estimate

$$\int_{F^{-1}(B_0)} \sigma_1^2 d\mu \geq c_d \sqrt{\frac{S_3}{S_2}} \mu(F^{-1}(B_0))$$

for some constant $c_d > 0$, then

$$R_1 R_2 R_3 R_4 \geq \frac{c_d}{9} S_1 S_2^{1/2} S_3^{3/2} S_4.$$

That is, the argument proves that the problem is solvable given a substantial additional assumption.

Overall, the argument is a little verbose, but it seems to be essentially correct. There are two main problems: first, as noted above, the argument needs an additional assumption. Second, in the original problem, there are two cases: that $R_3 R_4 > k S_3 S_4$ or that $R_1 R_2 R_3 R_4 > k S_1 S_2^{1/2} S_3^{3/2} S_4$. The argument aims to show that if the additional condition holds, then the second inequality is true, but that leaves out one of the cases.

This is a strange submission. It spends most of its space proving that

$$\text{FillVol}_R(z_y) \geq \text{FillVol}_S(P_y) = S_1 S_2 \text{ dist}(y, \partial B).$$

This is an intermediate result that shows up in other submissions and in the problem solution, but this submission doesn’t use it to prove anything – it just states it and moves on.

Novelty.

2 Recommendation

Reject

3 Submission with inline comments

Theorem (fiber-filling and coarea reduction)

Let

$$R = \prod_{i=1}^4 [0, R_i], \quad S = \prod_{i=1}^4 [0, S_i],$$

with $0 < R_1 \leq R_2 \leq R_3 \leq R_4$ and $0 < S_1 \leq S_2 \leq S_3 \leq S_4$. Orient S by

$$dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4.$$

Write

$$Q = [0, S_1] \times [0, S_2], \quad B = [0, S_3] \times [0, S_4],$$

so $S = Q \times B$, and orient Q and B by $dx_1 \wedge dx_2$ and $dx_3 \wedge dx_4$, respectively. Let

$$\pi : S \rightarrow B, \quad \pi(x_1, x_2, x_3, x_4) = (x_3, x_4),$$

and suppose

$$f : (R, \partial R) \rightarrow (S, \partial S)$$

is a Lipschitz finite piecewise smooth map of degree 1 with $\text{Dil}_2(f) \leq 1$. Put

$$F = \pi \circ f = (f_3, f_4), \quad G = (f_1, f_2).$$

For $y \in B$, set

$$P_y = Q \times \{y\}.$$

Use the slicing convention normalized by

$$\langle \llbracket Q \times B \rrbracket, \pi, y \rangle = \llbracket Q \times \{y\} \rrbracket.$$

Then for a.e. $y \in \text{int } B$, the slice

$$z_y = \langle \llbracket R \rrbracket, F, y \rangle$$

is a relative integral 2-cycle in $(R, \partial R)$, and

$$f_{\#} z_y = \llbracket P_y \rrbracket.$$

Consequently,

$$\text{FillVol}_R(z_y) \geq \text{FillVol}_S(P_y) = S_1 S_2 \text{dist}(y, \partial B).$$

Moreover,

$$\int_B \text{Area}(z_y) dy \leq \text{Vol}(R).$$

Finally, let

$$B_0 = \left[\frac{S_3}{3}, \frac{2S_3}{3} \right] \times \left[\frac{S_4}{3}, \frac{2S_4}{3} \right].$$

At points where dF has rank 2, let

$$\sigma_1(x) \geq \sigma_2(x) > 0$$

be the two nonzero singular values of dF_x . Define the measure

$$d\mu = \frac{\sigma_2}{\sigma_1} dx$$

on the rank-two set of dF , and $d\mu = 0$ where $\text{rank } dF < 2$. Then

$$\mu(F^{-1}(B_0)) \geq \frac{1}{9} S_1 S_2 S_3 S_4$$

and

$$\int_{B_0} \text{Area}(z_y) dy = \int_{F^{-1}(B_0)} \sigma_1^2 d\mu.$$

Thus, if in addition one has an estimate

$$\int_{F^{-1}(B_0)} \sigma_1^2 d\mu \geq c_d \sqrt{\frac{S_3}{S_2}} \mu(F^{-1}(B_0))$$

for some constant $c_d > 0$, then

$$R_1 R_2 R_3 R_4 \geq \frac{c_d}{9} S_1 S_2^{1/2} S_3^{3/2} S_4.$$

Proof

Write $\mathbf{M}(T)$ for the mass of an integral current T , and write $\llbracket X \rrbracket$ for the integral current associated to an oriented Lipschitz manifold or rectangle X .

For a compact rectangle X , if $Z \in \mathbf{I}_m(X)$ satisfies $\text{spt } \partial Z \subset \partial X$, define its relative filling volume by

$$\text{FillVol}_X(Z) = \inf \{ \mathbf{M}(Y) : Y \in \mathbf{I}_{m+1}(X), \text{spt}(\partial Y - Z) \subset \partial X \}.$$

We use the standard slicing theorem for integral currents: if $T \in \mathbf{I}_m$ and Φ is Lipschitz into $\mathbb{R}^k$, then for a.e. a , the slice $\langle T, \Phi, a \rangle \in \mathbf{I}_{m-k}$ exists and

$$\partial \langle T, \Phi, a \rangle = \pm \langle \partial T, \Phi, a \rangle.$$

We also use slicing naturality: for Lipschitz H ,

$$H_{\#} \langle T, \Phi \circ H, a \rangle = \langle H_{\#} T, \Phi, a \rangle$$

for a.e. a . Finally, we use the rectifiable-current area inequality

$$\mathbf{M}(H_{\#} T) \leq \int J_k(dH|_{\text{Tan } T}) d\|T\|$$

for $T \in \mathbf{I}_k$.

First prove the fundamental current identity

$$f_{\#} \llbracket R \rrbracket = \llbracket S \rrbracket.$$

Let

$$T = f_{\#} \llbracket R \rrbracket.$$

Since f is Lipschitz, $T \in \mathbf{I}_4(S)$, and

$$\partial T = f_{\#} \partial \llbracket R \rrbracket$$

is supported in $f(\partial R) \subset \partial S$. Hence, in $\text{int } S$,

$$\partial(T \llcorner \text{int } S) = 0.$$

By the constancy theorem on the connected open set $\text{int } S$,

$$T \llcorner \text{int } S = m \llbracket \text{int } S \rrbracket$$

for some $m \in \mathbb{Z}$. Therefore $T - m \llbracket S \rrbracket$ is supported in ∂S . Since ∂S has $\mathcal{H}^4$ -measure zero, no nonzero integral 4-current can be supported there, so

$$T = m \llbracket S \rrbracket.$$

Because f has degree 1, T represents the relative fundamental class $[S, \partial S]$, hence $m = 1$. Thus

$$f_{\#} \llbracket R \rrbracket = \llbracket S \rrbracket.$$

Now apply slicing to $F = \pi \circ f$. For a.e. $y \in \text{int } B$,

$$z_y = \langle \llbracket R \rrbracket, F, y \rangle$$

is an integral 2-current, and

$$\partial z_y = \pm \langle \partial \llbracket R \rrbracket, F, y \rangle,$$

so ∂z_y is supported in ∂R . Hence z_y is a relative integral 2-cycle in $(R, \partial R)$.

By slicing naturality,

$$f_{\#} z_y = f_{\#} \langle \llbracket R \rrbracket, F, y \rangle = \langle f_{\#} \llbracket R \rrbracket, \pi, y \rangle = \langle \llbracket S \rrbracket, \pi, y \rangle.$$

With the stated product orientation and slice convention,

$$\langle \llbracket S \rrbracket, \pi, y \rangle = \llbracket Q \times \{y\} \rrbracket = \llbracket P_y \rrbracket.$$

Thus

$$f_{\#} z_y = \llbracket P_y \rrbracket.$$

Next, $\text{Dil}_2(f) \leq 1$ implies $\text{Dil}_3(f) \leq 1$. Indeed, at a differentiability point of f , let

$$a_1 \geq a_2 \geq a_3 \geq a_4 \geq 0$$

be the singular values of df . The hypothesis gives $a_1 a_2 \leq 1$. Hence $a_2 \leq 1$, so $a_3 \leq 1$, and therefore

$$a_1 a_2 a_3 \leq a_1 a_2 \leq 1.$$

Thus $\|\Lambda^3 df\| \leq 1$ a.e.

Let $Y \in \mathbf{I}_3(R)$ be a relative filling of z_y , so

$$\text{spt}(\partial Y - z_y) \subset \partial R.$$

Then

$$\partial(f_{\#}Y) - f_{\#}z_y = f_{\#}(\partial Y - z_y)$$

is supported in $f(\partial R) \subset \partial S$. Since $f_{\#}z_y = \llbracket P_y \rrbracket$, the current $f_{\#}Y$ is a relative filling of P_y in S . Also,

$$\mathbf{M}(f_{\#}Y) \leq \int \|\Lambda^3 df\| d\|Y\| \leq \mathbf{M}(Y).$$

Taking the infimum over all such Y gives

$$\text{FillVol}_R(z_y) \geq \text{FillVol}_S(P_y).$$

It remains to compute $\text{FillVol}_S(P_y)$. Let

$$\delta(y) = \text{dist}(y, \partial B).$$

For the upper bound, choose a shortest segment $\gamma \subset B$ from ∂B to y , oriented toward y . Then

$$C = \llbracket Q \rrbracket \times \llbracket \gamma \rrbracket$$

is a relative filling of P_y in S , and

$$\mathbf{M}(C) = S_1 S_2 \delta(y).$$

Hence

$$\text{FillVol}_S(P_y) \leq S_1 S_2 \delta(y).$$

For the lower bound, define

$$\delta(q) = \text{dist}(q, \partial B) = \min\{q_3, S_3 - q_3, q_4, S_4 - q_4\}.$$

For $\eta > 0$, set

$$m_{\eta}(q) = -\eta \log \left(e^{-q_3/\eta} + e^{-(S_3 - q_3)/\eta} + e^{-q_4/\eta} + e^{-(S_4 - q_4)/\eta} \right).$$

Then

$$\delta(q) - \eta \log 4 \leq m_{\eta}(q) \leq \delta(q), \quad |\nabla m_{\eta}(q)| \leq 1.$$

Choose a smooth function $\chi_{\eta} : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\chi_{\eta}(t) = 0 \quad \text{for } t \leq 0, \quad 0 \leq \chi'_{\eta}(t) \leq 1, \quad \chi_{\eta}(t) \geq t - \eta \quad \text{for } t \geq 0.$$

Set

$$\rho_\eta(q) = \chi_\eta(m_\eta(q)).$$

Then ρ_η is smooth,

$$\rho_\eta = 0 \quad \text{on } \partial B, \quad |d\rho_\eta| \leq 1,$$

and for every fixed $y \in \text{int } B$,

$$\rho_\eta(y) \rightarrow \delta(y) \quad \text{as } \eta \rightarrow 0.$$

Define the smooth 2-form

$$\alpha_\eta = \rho_\eta(x_3, x_4) dx_1 \wedge dx_2.$$

It restricts to zero on ∂S : on the x_1 - and x_2 -faces because $dx_1 \wedge dx_2$ vanishes there, and on the x_3 - and x_4 -faces because $\rho_\eta = 0$ on ∂B .

Let $Y \in \mathbf{I}_3(S)$ be any relative filling of P_y , so

$$\text{spt}(\partial Y - \llbracket P_y \rrbracket) \subset \partial S.$$

Since α_η vanishes on ∂S ,

$$\partial Y(\alpha_\eta) = \llbracket P_y \rrbracket(\alpha_\eta) = \rho_\eta(y) S_1 S_2.$$

By Stokes' theorem for currents,

$$Y(d\alpha_\eta) = \rho_\eta(y) S_1 S_2.$$

But

$$d\alpha_\eta = d\rho_\eta \wedge dx_1 \wedge dx_2,$$

whose comass is at most $|d\rho_\eta| \leq 1$. Therefore

$$\mathbf{M}(Y) \geq \rho_\eta(y) S_1 S_2.$$

Letting $\eta \rightarrow 0$ gives

$$\mathbf{M}(Y) \geq S_1 S_2 \delta(y).$$

Taking the infimum over Y ,

$$\text{FillVol}_S(P_y) \geq S_1 S_2 \delta(y).$$

Thus

$$\text{FillVol}_S(P_y) = S_1 S_2 \text{dist}(y, \partial B).$$

In particular, for $y \in B_0$,

$$\text{dist}(y, \partial B) \geq \frac{S_3}{3},$$

because $S_4 \geq S_3$. Hence

$$\text{FillVol}_R(z_y) \geq \frac{1}{3} S_1 S_2 S_3.$$

Reviewer 2

In most arguments, the next step would be to use the relative isoperimetric inequality to prove a lower bound on $\text{Area}(z_y)$, but this is the last place that this argument mentions FillVol .

Now apply the coarea formula to the Lipschitz map $F : R \rightarrow B$. For every nonnegative Borel function g ,

$$\int_B \int_{F^{-1}(y)} g \, d\mathcal{H}^2 \, dy = \int_R g \, J_2 F \, dx,$$

where

$$J_2 F = \|\Lambda^2 dF\|.$$

For a.e. y , the mass of the slice z_y is $\text{Area}(z_y)$, so with $g = 1$,

$$\int_B \text{Area}(z_y) \, dy = \int_R J_2 F \, dx.$$

Since $F = \pi \circ f$ and π is an orthogonal projection,

$$J_2 F \leq \|\Lambda^2 df\| \leq 1.$$

Therefore

$$\int_B \text{Area}(z_y) \, dy \leq \text{Vol}(R).$$

It remains to prove the two estimates involving μ . Fix a regular value $y \in B_0$, and let $x \in z_y$ be a point where f is differentiable and dF_x has rank 2. Then

$$T_x z_y = \ker dF_x.$$

Let

$$N_x = (T_x z_y)^\perp.$$

The restriction $dF_x|_{N_x} : N_x \rightarrow \mathbb{R}^2$ has singular values $\sigma_1(x) \geq \sigma_2(x) > 0$.

Choose a unit vector $n \in N_x$ such that

$$|dF_x(n)| = \sigma_1(x).$$

For any unit vector $v \in T_x z_y$, one has $dF_x(v) = 0$. Since

$$df_x = (dG_x, dF_x),$$

we get

$$df_x(v) = (dG_x(v), 0), \quad df_x(n) = (dG_x(n), dF_x(n)).$$

In the orthogonal splitting $\mathbb{R}^4 = \mathbb{R}_Q^2 \oplus \mathbb{R}_B^2$, the mixed $Q \wedge B$ component of

$$df_x(v) \wedge df_x(n)$$

is

$$dG_x(v) \wedge dF_x(n),$$

whose norm is

$$|dG_x(v)| |dF_x(n)| = |dG_x(v)| \sigma_1(x).$$

This mixed component has norm at most the norm of the full wedge. Therefore, using $\text{Dil}_2(f) \leq 1$,

$$|dG_x(v)| \sigma_1(x) \leq |df_x(v) \wedge df_x(n)| \leq 1.$$

Hence

$$|dG_x(v)| \leq \sigma_1(x)^{-1}.$$

Taking the supremum over unit $v \in T_x z_y$,

$$\|dG_x|_{T_x z_y}\| \leq \sigma_1(x)^{-1}.$$

Since $T_x z_y$ is two-dimensional,

$$J_2(G|_{T_x z_y}) \leq \sigma_1(x)^{-2}.$$

Let $p_Q : S \rightarrow Q$ be the projection onto the first two coordinates. Since $G = p_Q \circ f$, for a.e. $y \in B_0$,

$$G_{\#} z_y = (p_Q)_{\#} f_{\#} z_y = (p_Q)_{\#} \llbracket P_y \rrbracket = \llbracket Q \rrbracket.$$

Therefore

$$S_1 S_2 = \mathbf{M}(\llbracket Q \rrbracket) = \mathbf{M}(G_{\#} z_y) \leq \int_{z_y} J_2(G|_{T_x z_y}) dA \leq \int_{z_y} \sigma_1^{-2} dA.$$

Integrating over B_0 ,

$$\int_{B_0} \int_{z_y} \sigma_1^{-2} dA dy \geq S_1 S_2 |B_0| = \frac{1}{9} S_1 S_2 S_3 S_4.$$

Now define

$$\tilde{g}(x) = \sigma_1(x)^{-2} \mathbf{1}_{F^{-1}(B_0) \cap \{\text{rank } dF=2\}}(x).$$

For $M > 0$, set $g_M = \min\{\tilde{g}, M\}$. By the coarea formula,

$$\int_{B_0} \int_{z_y} g_M dA dy = \int_{F^{-1}(B_0)} g_M J_2 F dx.$$

Letting $M \rightarrow \infty$ and using monotone convergence,

$$\int_{B_0} \int_{z_y} \sigma_1^{-2} dA dy = \int_{F^{-1}(B_0) \cap \{\text{rank } dF=2\}} \sigma_1^{-2} J_2 F dx.$$

On the rank-two set,

$$J_2 F = \sigma_1 \sigma_2.$$

Thus

$$\sigma_1^{-2} J_2 F = \frac{\sigma_2}{\sigma_1}.$$

Hence

$$\int_{B_0} \int_{z_y} \sigma_1^{-2} dA dy = \int_{F^{-1}(B_0)} \frac{\sigma_2}{\sigma_1} dx = \mu(F^{-1}(B_0)).$$

Combining this with the previous inequality gives

$$\mu(F^{-1}(B_0)) \geq \frac{1}{9} S_1 S_2 S_3 S_4.$$

Finally, applying the ordinary coarea formula over B_0 ,

$$\int_{B_0} \text{Area}(z_y) dy = \int_{F^{-1}(B_0)} J_2 F dx.$$

On the rank-two set,

$$J_2 F = \sigma_1 \sigma_2 = \sigma_1^2 \frac{\sigma_2}{\sigma_1},$$

while both sides vanish on the rank < 2 set. Therefore

$$\int_{B_0} \text{Area}(z_y) dy = \int_{F^{-1}(B_0)} \sigma_1^2 d\mu.$$

Assume now that

$$\int_{F^{-1}(B_0)} \sigma_1^2 d\mu \geq c_d \sqrt{\frac{S_3}{S_2}} \mu(F^{-1}(B_0)).$$

Then

$$\int_{B_0} \text{Area}(z_y) dy \geq c_d \sqrt{\frac{S_3}{S_2}} \mu(F^{-1}(B_0)) \geq \frac{c_d}{9} S_1 S_2 S_3 S_4 \sqrt{\frac{S_3}{S_2}}.$$

Since

$$\text{Vol}(R) \geq \int_B \text{Area}(z_y) dy \geq \int_{B_0} \text{Area}(z_y) dy,$$

we obtain

$$R_1 R_2 R_3 R_4 = \text{Vol}(R) \geq \frac{c_d}{9} S_1 S_2^{1/2} S_3^{3/2} S_4.$$

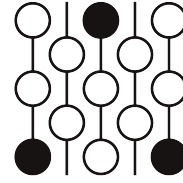
This completes the proof.

Review of Submission 4B

First Proof Second Batch

Reviewer 3

June 4–8, 2026



1 Review

Correctness. The proofs of what it claims to prove are correct, but it only solves the whole problem conditional on an odd-looking estimate without a clear idea behind it and that might or might not be correct, and if it is correct might have essentially the same difficulty as the original problem.

Novelty. In terms of novelty, it is mostly just using the coarea formula together with a little bit of linear algebra in places. There does not appear to be new ideas here, except if coming up with the odd-looking estimate above counts as that. It discusses and proves some filling inequalities which are also fairly standard (see statement of the main theorem and pg. 6) but does not seem to use them for anything.

Submission A's solution had some common features with this one (reducing everything to a brain-in-a-jar estimate whose truth seemed hard to determine), but was able to actually get something out of the filling inequalities it proved and to obtain statements that resembled the problem conclusion before it went off the rails.

2 Recommendation

I would recommend to reject.

3 Submission with inline comments

Theorem (fiber-filling and coarea reduction)

Let

$$R = \prod_{i=1}^4 [0, R_i], \quad S = \prod_{i=1}^4 [0, S_i],$$

with $0 < R_1 \leq R_2 \leq R_3 \leq R_4$ and $0 < S_1 \leq S_2 \leq S_3 \leq S_4$. Orient S by

$$dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4.$$

Write

$$Q = [0, S_1] \times [0, S_2], \quad B = [0, S_3] \times [0, S_4],$$

so $S = Q \times B$, and orient Q and B by $dx_1 \wedge dx_2$ and $dx_3 \wedge dx_4$, respectively. Let

$$\pi : S \rightarrow B, \quad \pi(x_1, x_2, x_3, x_4) = (x_3, x_4),$$

and suppose

$$f : (R, \partial R) \rightarrow (S, \partial S)$$

is a Lipschitz finite piecewise smooth map of degree 1 with $\text{Dil}_2(f) \leq 1$. Put

$$F = \pi \circ f = (f_3, f_4), \quad G = (f_1, f_2).$$

For $y \in B$, set

$$P_y = Q \times \{y\}.$$

Use the slicing convention normalized by

$$\langle \llbracket Q \times B \rrbracket, \pi, y \rangle = \llbracket Q \times \{y\} \rrbracket.$$

Then for a.e. $y \in \text{int } B$, the slice

$$z_y = \langle \llbracket R \rrbracket, F, y \rangle$$

is a relative integral 2-cycle in $(R, \partial R)$, and

$$f_{\#} z_y = \llbracket P_y \rrbracket.$$

Consequently,

$$\text{FillVol}_R(z_y) \geq \text{FillVol}_S(P_y) = S_1 S_2 \text{dist}(y, \partial B).$$

Moreover,

$$\int_B \text{Area}(z_y) dy \leq \text{Vol}(R).$$

Finally, let

$$B_0 = \left[\frac{S_3}{3}, \frac{2S_3}{3} \right] \times \left[\frac{S_4}{3}, \frac{2S_4}{3} \right].$$

At points where dF has rank 2, let

$$\sigma_1(x) \geq \sigma_2(x) > 0$$

be the two nonzero singular values of dF_x . Define the measure

$$d\mu = \frac{\sigma_2}{\sigma_1} dx$$

on the rank-two set of dF , and $d\mu = 0$ where $\text{rank } dF < 2$. Then

$$\mu(F^{-1}(B_0)) \geq \frac{1}{9} S_1 S_2 S_3 S_4$$

and

$$\int_{B_0} \text{Area}(z_y) dy = \int_{F^{-1}(B_0)} \sigma_1^2 d\mu.$$

Thus, if in addition one has an estimate

$$\int_{F^{-1}(B_0)} \sigma_1^2 d\mu \geq c_d \sqrt{\frac{S_3}{S_2}} \mu(F^{-1}(B_0))$$

for some constant $c_d > 0$, then

$$R_1 R_2 R_3 R_4 \geq \frac{c_d}{9} S_1 S_2^{1/2} S_3^{3/2} S_4.$$

Reviewer 3

The theorem statement was written in a strange way. It seemed to be presenting the argument in general terms, but at least on a first reading it was unclear which steps were meant to be obvious and not to require justification, and which steps were to be justified later, in the proof below.

Proof

Write $\mathbf{M}(T)$ for the mass of an integral current T , and write $\llbracket X \rrbracket$ for the integral current associated to an oriented Lipschitz manifold or rectangle X .

For a compact rectangle X , if $Z \in \mathbf{I}_m(X)$ satisfies $\text{spt } \partial Z \subset \partial X$, define its relative filling volume by

$$\text{FillVol}_X(Z) = \inf \{ \mathbf{M}(Y) : Y \in \mathbf{I}_{m+1}(X), \text{spt}(\partial Y - Z) \subset \partial X \}.$$

We use the standard slicing theorem for integral currents: if $T \in \mathbf{I}_m$ and Φ is Lipschitz into $\mathbb{R}^k$, then for a.e. a , the slice $\langle T, \Phi, a \rangle \in \mathbf{I}_{m-k}$ exists and

$$\partial \langle T, \Phi, a \rangle = \pm \langle \partial T, \Phi, a \rangle.$$

We also use slicing naturality: for Lipschitz H ,

$$H_{\#} \langle T, \Phi \circ H, a \rangle = \langle H_{\#} T, \Phi, a \rangle$$

for a.e. a . Finally, we use the rectifiable-current area inequality

$$\mathbf{M}(H_{\#} T) \leq \int J_k(dH|_{\text{Tan } T}) d\|T\|$$

for $T \in \mathbf{I}_k$.

First prove the fundamental current identity

$$f_{\#} \llbracket R \rrbracket = \llbracket S \rrbracket.$$

Let

$$T = f_{\#} \llbracket R \rrbracket.$$

Since f is Lipschitz, $T \in \mathbf{I}_4(S)$, and

$$\partial T = f_{\#} \partial \llbracket R \rrbracket$$

is supported in $f(\partial R) \subset \partial S$. Hence, in $\text{int } S$,

$$\partial(T \llcorner \text{int } S) = 0.$$

By the constancy theorem on the connected open set $\text{int } S$,

$$T \llcorner \text{int } S = m \llbracket \text{int } S \rrbracket$$

for some $m \in \mathbb{Z}$. Therefore $T - m \llbracket S \rrbracket$ is supported in ∂S . Since ∂S has $\mathcal{H}^4$ -measure zero, no nonzero integral 4-current can be supported there, so

$$T = m \llbracket S \rrbracket.$$

Because f has degree 1, T represents the relative fundamental class $[S, \partial S]$, hence $m = 1$. Thus

$$f_{\#} \llbracket R \rrbracket = \llbracket S \rrbracket.$$

Now apply slicing to $F = \pi \circ f$. For a.e. $y \in \text{int } B$,

$$z_y = \langle \llbracket R \rrbracket, F, y \rangle$$

is an integral 2-current, and

$$\partial z_y = \pm \langle \partial \llbracket R \rrbracket, F, y \rangle,$$

so ∂z_y is supported in ∂R . Hence z_y is a relative integral 2-cycle in $(R, \partial R)$.

By slicing naturality,

$$f_{\#} z_y = f_{\#} \langle \llbracket R \rrbracket, F, y \rangle = \langle f_{\#} \llbracket R \rrbracket, \pi, y \rangle = \langle \llbracket S \rrbracket, \pi, y \rangle.$$

With the stated product orientation and slice convention,

$$\langle \llbracket S \rrbracket, \pi, y \rangle = \llbracket Q \times \{y\} \rrbracket = \llbracket P_y \rrbracket.$$

Thus

$$f_{\#} z_y = \llbracket P_y \rrbracket.$$

Next, $\text{Dil}_2(f) \leq 1$ implies $\text{Dil}_3(f) \leq 1$. Indeed, at a differentiability point of f , let

$$a_1 \geq a_2 \geq a_3 \geq a_4 \geq 0$$

be the singular values of df . The hypothesis gives $a_1 a_2 \leq 1$. Hence $a_2 \leq 1$, so $a_3 \leq 1$, and therefore

$$a_1 a_2 a_3 \leq a_1 a_2 \leq 1.$$

Thus $\|\Lambda^3 df\| \leq 1$ a.e.

Let $Y \in \mathbf{I}_3(R)$ be a relative filling of z_y , so

$$\text{spt}(\partial Y - z_y) \subset \partial R.$$

Then

$$\partial(f_{\#}Y) - f_{\#}z_y = f_{\#}(\partial Y - z_y)$$

is supported in $f(\partial R) \subset \partial S$. Since $f_{\#}z_y = \llbracket P_y \rrbracket$, the current $f_{\#}Y$ is a relative filling of P_y in S . Also,

$$\mathbf{M}(f_{\#}Y) \leq \int \|\Lambda^3 df\| d\|Y\| \leq \mathbf{M}(Y).$$

Taking the infimum over all such Y gives

$$\text{FillVol}_R(z_y) \geq \text{FillVol}_S(P_y).$$

It remains to compute $\text{FillVol}_S(P_y)$. Let

$$\delta(y) = \text{dist}(y, \partial B).$$

For the upper bound, choose a shortest segment $\gamma \subset B$ from ∂B to y , oriented toward y . Then

$$C = \llbracket Q \rrbracket \times \llbracket \gamma \rrbracket$$

is a relative filling of P_y in S , and

$$\mathbf{M}(C) = S_1 S_2 \delta(y).$$

Hence

$$\text{FillVol}_S(P_y) \leq S_1 S_2 \delta(y).$$

For the lower bound, define

Reviewer 3

The proof of the lower bound seems correct but is unnecessarily complicated. (It isn't actually used, neither are any of the other statements about filling inequalities.)

$$\delta(q) = \text{dist}(q, \partial B) = \min\{q_3, S_3 - q_3, q_4, S_4 - q_4\}.$$

For $\eta > 0$, set

$$m_{\eta}(q) = -\eta \log \left(e^{-q_3/\eta} + e^{-(S_3-q_3)/\eta} + e^{-q_4/\eta} + e^{-(S_4-q_4)/\eta} \right).$$

Then

$$\delta(q) - \eta \log 4 \leq m_{\eta}(q) \leq \delta(q), \quad |\nabla m_{\eta}(q)| \leq 1.$$

Choose a smooth function $\chi_{\eta} : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\chi_{\eta}(t) = 0 \quad \text{for } t \leq 0, \quad 0 \leq \chi'_{\eta}(t) \leq 1, \quad \chi_{\eta}(t) \geq t - \eta \quad \text{for } t \geq 0.$$

Set

$$\rho_{\eta}(q) = \chi_{\eta}(m_{\eta}(q)).$$

Then ρ_η is smooth,

$$\rho_\eta = 0 \quad \text{on } \partial B, \quad |d\rho_\eta| \leq 1,$$

and for every fixed $y \in \text{int } B$,

$$\rho_\eta(y) \rightarrow \delta(y) \quad \text{as } \eta \rightarrow 0.$$

Define the smooth 2-form

$$\alpha_\eta = \rho_\eta(x_3, x_4) dx_1 \wedge dx_2.$$

It restricts to zero on ∂S : on the x_1 - and x_2 -faces because $dx_1 \wedge dx_2$ vanishes there

$dx_1 \wedge dx_2$ does not vanish on the x_1 and x_2 -faces

Reviewer 3

, and on the x_3 - and x_4 -faces because $\rho_\eta = 0$ on ∂B .

Let $Y \in \mathbf{I}_3(S)$ be any relative filling of P_y , so

$$\text{spt}(\partial Y - \llbracket P_y \rrbracket) \subset \partial S.$$

Since α_η vanishes on ∂S ,

$$\partial Y(\alpha_\eta) = \llbracket P_y \rrbracket(\alpha_\eta) = \rho_\eta(y) S_1 S_2.$$

By Stokes' theorem for currents,

$$Y(d\alpha_\eta) = \rho_\eta(y) S_1 S_2.$$

But

$$d\alpha_\eta = d\rho_\eta \wedge dx_1 \wedge dx_2,$$

whose comass is at most $|d\rho_\eta| \leq 1$. Therefore

$$\mathbf{M}(Y) \geq \rho_\eta(y) S_1 S_2.$$

Letting $\eta \rightarrow 0$ gives

$$\mathbf{M}(Y) \geq S_1 S_2 \delta(y).$$

Taking the infimum over Y ,

$$\text{FillVol}_S(P_y) \geq S_1 S_2 \delta(y).$$

Thus

$$\text{FillVol}_S(P_y) = S_1 S_2 \text{dist}(y, \partial B).$$

In particular, for $y \in B_0$,

$$\text{dist}(y, \partial B) \geq \frac{S_3}{3},$$

because $S_4 \geq S_3$. Hence

$$\text{FillVol}_R(z_y) \geq \frac{1}{3} S_1 S_2 S_3.$$

It doesn't seem to use this filling estimate at all, or the filling estimate mentioned earlier in the statement of the theorem.

Reviewer 3

Now apply the coarea formula to the Lipschitz map $F : R \rightarrow B$. For every nonnegative Borel function g ,

$$\int_B \int_{F^{-1}(y)} g \, d\mathcal{H}^2 \, dy = \int_R g \, J_2 F \, dx,$$

where

$$J_2 F = \|\Lambda^2 dF\|.$$

For a.e. y , the mass of the slice z_y is $\text{Area}(z_y)$, so with $g = 1$,

$$\int_B \text{Area}(z_y) \, dy = \int_R J_2 F \, dx.$$

Since $F = \pi \circ f$ and π is an orthogonal projection,

$$J_2 F \leq \|\Lambda^2 df\| \leq 1.$$

Therefore

$$\int_B \text{Area}(z_y) \, dy \leq \text{Vol}(R).$$

It remains to prove the two estimates involving μ . Fix a regular value $y \in B_0$, and let $x \in z_y$ be a point where f is differentiable and dF_x has rank 2. Then

$$T_x z_y = \ker dF_x.$$

Let

$$N_x = (T_x z_y)^\perp.$$

The restriction $dF_x|_{N_x} : N_x \rightarrow \mathbb{R}^2$ has singular values $\sigma_1(x) \geq \sigma_2(x) > 0$.

Choose a unit vector $n \in N_x$ such that

$$|dF_x(n)| = \sigma_1(x).$$

For any unit vector $v \in T_x z_y$, one has $dF_x(v) = 0$. Since

$$df_x = (dG_x, dF_x),$$

we get

$$df_x(v) = (dG_x(v), 0), \quad df_x(n) = (dG_x(n), dF_x(n)).$$

In the orthogonal splitting $\mathbb{R}^4 = \mathbb{R}_Q^2 \oplus \mathbb{R}_B^2$, the mixed $Q \wedge B$ component of

$$df_x(v) \wedge df_x(n)$$

is

$$dG_x(v) \wedge dF_x(n),$$

whose norm is

$$|dG_x(v)| |dF_x(n)| = |dG_x(v)| \sigma_1(x).$$

This mixed component has norm at most the norm of the full wedge. Therefore, using $\text{Dil}_2(f) \leq 1$,

$$|dG_x(v)| \sigma_1(x) \leq |df_x(v) \wedge df_x(n)| \leq 1.$$

Hence

$$|dG_x(v)| \leq \sigma_1(x)^{-1}.$$

Taking the supremum over unit $v \in T_x z_y$,

$$\|dG_x|_{T_x z_y}\| \leq \sigma_1(x)^{-1}.$$

Since $T_x z_y$ is two-dimensional,

$$J_2(G|_{T_x z_y}) \leq \sigma_1(x)^{-2}.$$

Let $p_Q : S \rightarrow Q$ be the projection onto the first two coordinates. Since $G = p_Q \circ f$, for a.e. $y \in B_0$,

$$G_{\#} z_y = (p_Q)_{\#} f_{\#} z_y = (p_Q)_{\#} \llbracket P_y \rrbracket = \llbracket Q \rrbracket.$$

Therefore

$$S_1 S_2 = \mathbf{M}(\llbracket Q \rrbracket) = \mathbf{M}(G_{\#} z_y) \leq \int_{z_y} J_2(G|_{T_x z_y}) dA \leq \int_{z_y} \sigma_1^{-2} dA.$$

Integrating over B_0 ,

$$\int_{B_0} \int_{z_y} \sigma_1^{-2} dA dy \geq S_1 S_2 |B_0| = \frac{1}{9} S_1 S_2 S_3 S_4.$$

Now define

$$\tilde{g}(x) = \sigma_1(x)^{-2} \mathbf{1}_{F^{-1}(B_0) \cap \{\text{rank } dF=2\}}(x).$$

For $M > 0$, set $g_M = \min\{\tilde{g}, M\}$. By the coarea formula,

$$\int_{B_0} \int_{z_y} g_M dA dy = \int_{F^{-1}(B_0)} g_M J_2 F dx.$$

Letting $M \rightarrow \infty$ and using monotone convergence,

$$\int_{B_0} \int_{z_y} \sigma_1^{-2} dA dy = \int_{F^{-1}(B_0) \cap \{\text{rank } dF=2\}} \sigma_1^{-2} J_2 F dx.$$

On the rank-two set,

$$J_2 F = \sigma_1 \sigma_2.$$

Thus

$$\sigma_1^{-2} J_2 F = \frac{\sigma_2}{\sigma_1}.$$

Hence

Should say it's using the coarea formula for the next equality.

Reviewer 3

$$\int_{B_0} \int_{z_y} \sigma_1^{-2} dA dy = \int_{F^{-1}(B_0)} \frac{\sigma_2}{\sigma_1} dx = \mu(F^{-1}(B_0)).$$

Combining this with the previous inequality gives

$$\mu(F^{-1}(B_0)) \geq \frac{1}{9} S_1 S_2 S_3 S_4.$$

Finally, applying the ordinary coarea formula over B_0 ,

$$\int_{B_0} \text{Area}(z_y) dy = \int_{F^{-1}(B_0)} J_2 F dx.$$

On the rank-two set,

$$J_2 F = \sigma_1 \sigma_2 = \sigma_1^2 \frac{\sigma_2}{\sigma_1},$$

while both sides vanish on the rank < 2 set. Therefore

$$\int_{B_0} \text{Area}(z_y) dy = \int_{F^{-1}(B_0)} \sigma_1^2 d\mu.$$

Assume now that

$$\int_{F^{-1}(B_0)} \sigma_1^2 d\mu \geq c_d \sqrt{\frac{S_3}{S_2}} \mu(F^{-1}(B_0)).$$

Then

$$\int_{B_0} \text{Area}(z_y) dy \geq c_d \sqrt{\frac{S_3}{S_2}} \mu(F^{-1}(B_0)) \geq \frac{c_d}{9} S_1 S_2 S_3 S_4 \sqrt{\frac{S_3}{S_2}}.$$

Since

$$\text{Vol}(R) \geq \int_B \text{Area}(z_y) dy \geq \int_{B_0} \text{Area}(z_y) dy,$$

we obtain

$$R_1 R_2 R_3 R_4 = \text{Vol}(R) \geq \frac{c_d}{9} S_1 S_2^{1/2} S_3^{3/2} S_4.$$

This completes the proof.